STATISTICS

Hypothesis Testing Part 1

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Rev: 27.11.2023

Hypothesis Testing

Statistics

Hypothesis Testing

A hypothesis is an assertion about a parameter of a single population or two populations. There are two types of opposite statistical hypotheses:

1. Null Hypothesis (H_0)
2. Alternative Hypothesis (H_α) or (H_1)

Let θ be a population parameter and θ_0 be an asserted value of the parameter. Then, we can build three types of (H_0, H_a) hypothesis.

1)	$H_0: \theta = \theta_0$	2)	$H_0: \theta = \theta_0$ or $H_0: \theta \leq \theta_0$	3)	$H_0: \theta = \theta_0$ or $H_0: \theta \geq \theta_0$
	$H_a: \theta \neq \theta_0$		$H_a: \theta > \theta_0$		$H_a: \theta < \theta_0$

Two-Tailed Hypothesis

One-Tailed Hypothesis

Hypothesis Testing

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Hypothesis Testing

H_0 and H_a never intersect. H_0 always contains equality operator $(=, \leq, \geq)$, while H_a does not.

Example: It is claimed that the average life expectancy of women in Turkey is more than 72 years.

$H_0\colon \mu = 72$ or $\mu \leq 72$ ⇨ The average life expectancy of women in Turkey is not more than 72 years.

$H_a\colon \mu > 72$ ⇨ The average life expectancy of women in Turkey is more than 72 years.

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There are four possible situations in the process of hypothesis testing.

Decision	Exact Situation	
	H_0 is True	H_0 is False
Accept H_0	Correct Decision $(1 - \alpha)$	II.Type Error (β)
Reject H_0	I.Type Error (α)	Correct Decision $(1 - \beta)$

α : Significance level. It is the probability of rejecting H_0 when H_0 is true.

$1 - \alpha$: It states the confidence level of a test. It is the probability of accepting H_0 when H_0 is true.

$1 - \beta$: The power of the test. It is the probability of rejecting H_0 when H_0 is false.

Hypothesis Testing

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The Steps of Hypothesis Testing

The hypothesis testing is performed under the assumption that H_0 is correct. The steps of the process of hypothesis testing are given below:

1. Determine hypothesis H_0 and H_a.
2. Specify α.
3. Select the appropriate test statistic.
4. Select the appropriate decision rule. (Determine acceptance and rejection regions)
5. Compute the test statistics by using the sample.
6. Make a decision about (accept or reject) H_0.

Hypothesis Testing

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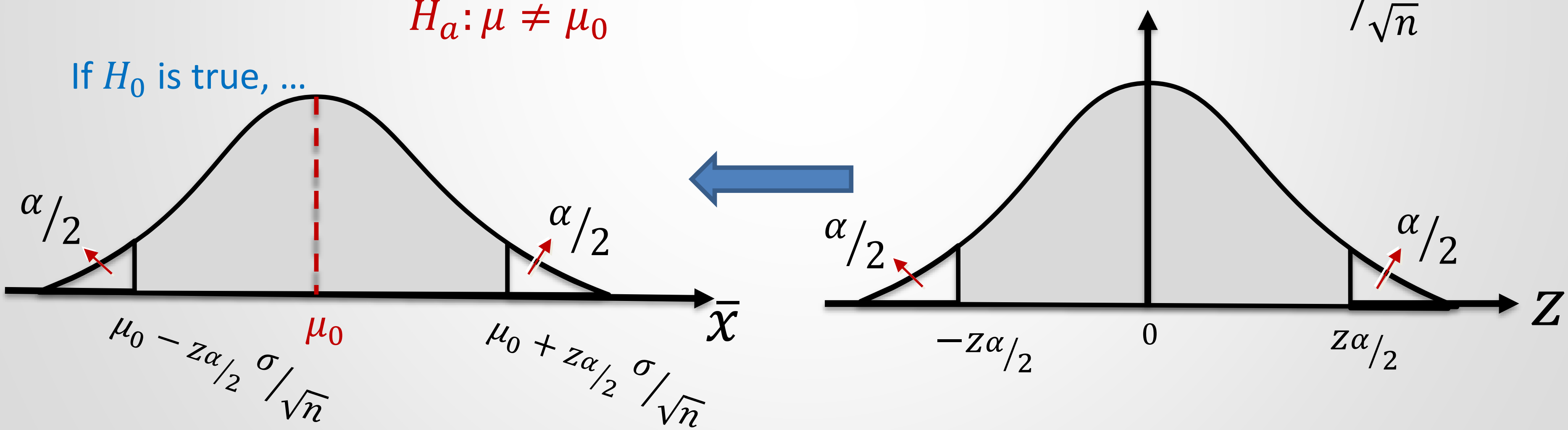
Hypothesis Testing for Population Mean μ:

A) For Normal Distributed Populations with Known Variance

1) Consider that $H_0: \mu = \mu_0$, $H_a: \mu \neq \mu_0$ for α.

Test Statistic: $Z_0 = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

If H_0 is true, ...



$$\mathrm{P}\left(\mu_0 - z_{\alpha/2}\,\sigma/\sqrt{n} \leq \bar{x} \leq \mu_0 + z_{\alpha/2}\,\sigma/\sqrt{n}\right) = 1 - \alpha \qquad \mathrm{P}\left(-z_{\alpha/2} \leq Z_0 \leq z_{\alpha/2}\right) = 1 - \alpha$$

The gray-shaded region is the acceptance region, while the other parts are rejection regions.

Hypothesis Testing

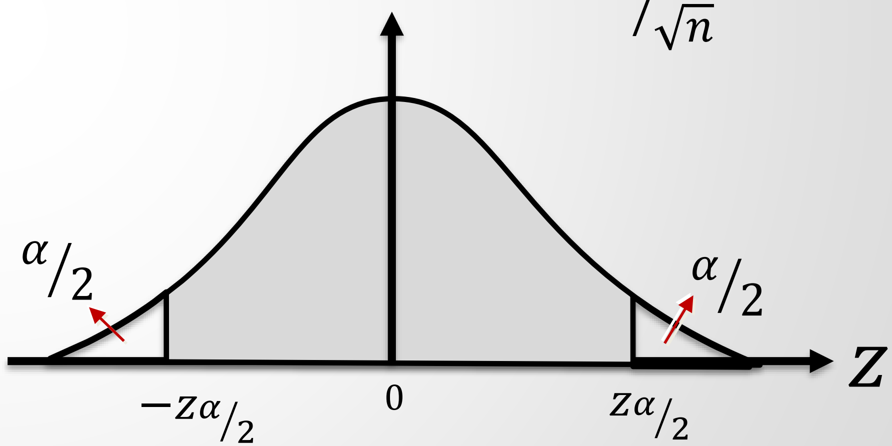
Statistics

Hypothesis Testing for Population Mean μ:

A) For Normal Distributed Populations with Known Variance

1) Consider that $H_0: \mu = \mu_0$, $H_a: \mu \neq \mu_0$ for α.

Test Statistic: $Z_0 = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$



Decision Rule:

If $-z\alpha_{/2} \leq Z_0 \leq z\alpha_{/2}$, accept H_0.
Otherwise, reject H_0.

$$P\left(-z\alpha_{/2} \leq Z_0 \leq z\alpha_{/2}\right) = 1 - \alpha$$

Hypothesis Testing

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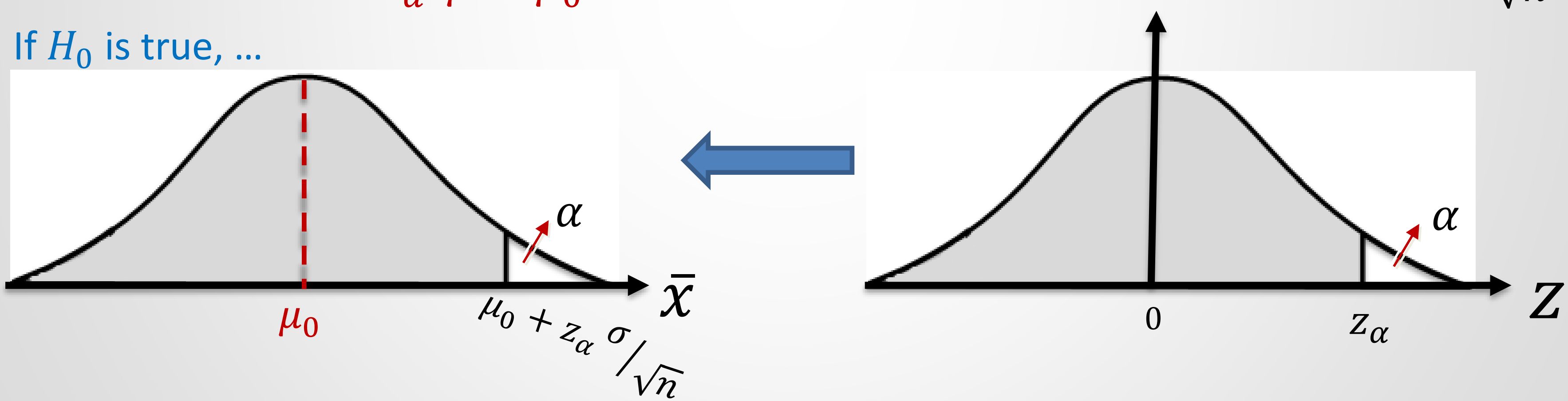
Hypothesis Testing for Population Mean μ:

A) For Normal Distributed Populations with Known Variance

2) Consider that $H_0\colon \mu = \mu_0$ or $\mu \leq \mu_0$
$H_a\colon \mu > \mu_0$ for α.

Test Statistic: $Z_0 = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

If H_0 is true, ...



$$P\left(\bar{x} \leq \mu_0 + z_\alpha \sigma/\sqrt{n}\right) = 1 - \alpha$$

$$P(Z_0 \leq z_\alpha) = 1 - \alpha$$

The gray shaded region is acceptance region, other part is rejection region.

Hypothesis Testing

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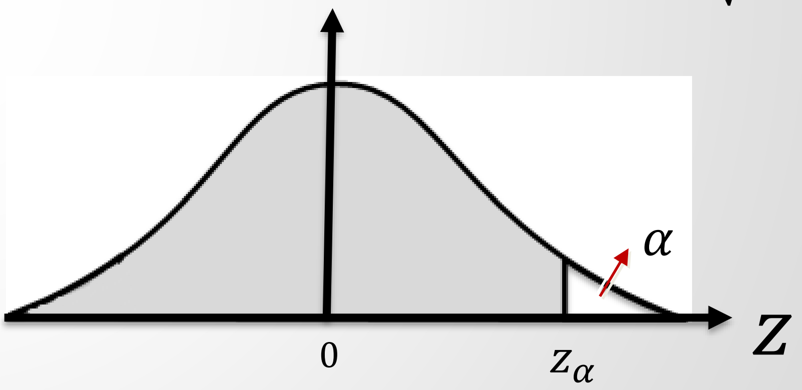
Hypothesis Testing for Population Mean μ:

A) For Normal Distributed Populations with Known Variance

2) Consider that $H_0: \mu = \mu_0$ or $\mu \leq \mu_0$ for α.
$H_a: \mu > \mu_0$

Test Statistic: $Z_0 = \dfrac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$



Decision Rule:

If $Z_0 \leq z_\alpha$, accept H_0.
Otherwise, reject H_0.

$$\mathrm{P}(Z_0 \leq z_\alpha) = 1 - \alpha$$

Hypothesis Testing

Statistics

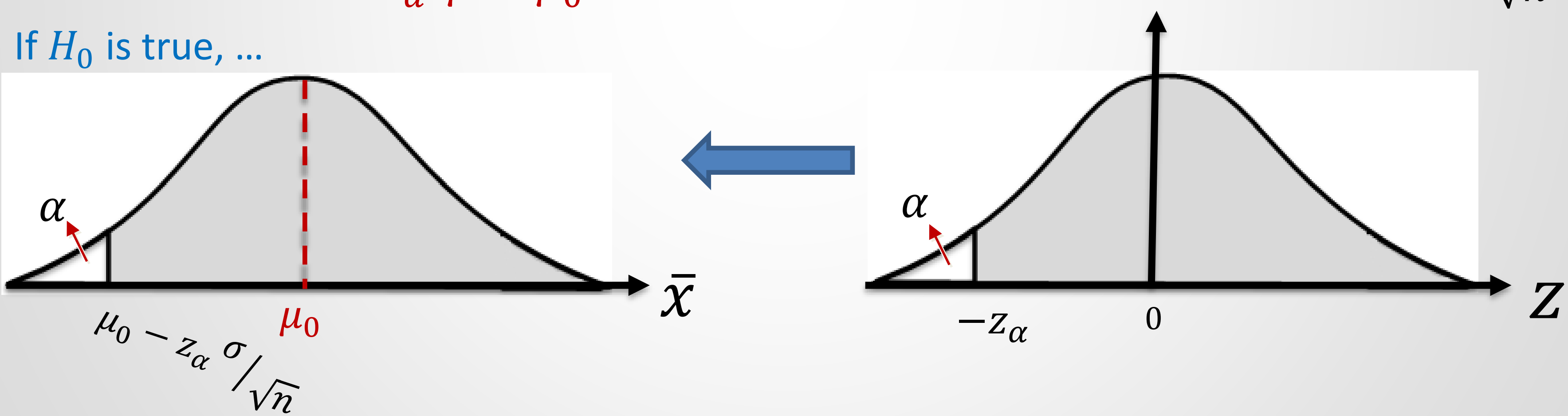
Hypothesis Testing for Population Mean μ:

A) For Normal Distributed Populations with Known Variance

3) Consider that $H_0: \mu = \mu_0$ or $\mu \geq \mu_0$ for α.
$H_a: \mu < \mu_0$

Test Statistic: $Z_0 = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

If H_0 is true, ...



$$\mathrm{P}\left(\mu_0 - z_\alpha \sigma/\sqrt{n} \leq \bar{x}\right) = 1 - \alpha$$

$$\mathrm{P}(-z_\alpha \leq Z_0) = 1 - \alpha$$

The gray shaded region is acceptance region, other part is rejection region.

Hypothesis Testing

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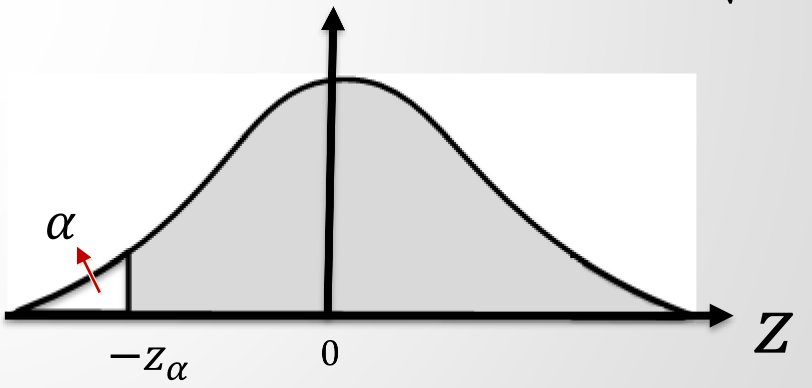
Hypothesis Testing for Population Mean μ:

A) For Normal Distributed Populations with Known Variance

3) Consider that $H_0: \mu = \mu_0$ or $\mu \geq \mu_0$ for α.
$H_a: \mu < \mu_0$

Test Statistic: $Z_0 = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$



Decision Rule:

If $-z_\alpha \leq Z_0$, accept H_0.
Otherwise, reject H_0.

$$P(-z_\alpha \leq Z_0) = 1 - \alpha$$

Hypothesis Testing

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Example 1: It is claimed that the IQ scores of undergraduate students in our country have a normal distribution with a mean of 117 and a variance of 160. The IQ scores of 12 randomly selected students have been measured to research this claim for the mean. The following scores are obtained:

103, 96, 112, 130, 111, 125, 89, 100, 104, 117, 99, 118.

Test whether the population mean is 117 or not ($\alpha = 0.05$).

Hypothesis Testing

Statistics

Hypothesis Testing for Population Mean μ:

We have investigated hypothesis testing of μ for the normally distributed population with known variance. Except for test statistic, the same analysis is valid for the populations with unknown variances and/or the populations with unknown distributions.

for normal distributed population with unknown variance $\Rightarrow$ Test Statistic: $t_0 = \dfrac{\bar{x} - \mu_0}{S/\sqrt{n}}$

for unknown distributed population with known variance ($n \geq 30$) $\Rightarrow$ Test Statistic: $z_0 = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

for unknown distributed population with unknown variance ($n \geq 30$) $\Rightarrow$ Test Statistic: $z_0 = \dfrac{\bar{x} - \mu_0}{S/\sqrt{n}}$

Hypothesis Testing

Statistics

Test Statistic	Hypothesis testing for the population mean (μ)
$z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	For the Normal distribution and known population variance
$z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	For the unknown distribution (n≥30) and known population variance
$z_0 = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}}$	For the unknown distribution (n≥30) and unknown population variance
Hypothesis	**Decision Rule**
1. $H_0: \mu = \mu_0$ $H_a: \mu \neq \mu_0$	If $z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$, then reject H_0.
2. $H_0: \mu = \mu_0$ or $\mu \geq \mu_0$ $H_a: \mu < \mu_0$	If $z_0 < -z_\alpha$, then reject H_0.
3. $H_0: \mu = \mu_0$ or $\mu \leq \mu_0$ $H_a: \mu > \mu_0$	If $z_0 > z_\alpha$, then reject H_0.
$t_0 = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}}$	For the Normal distribution and unknown population variance
Hypothesis	**Decision Rule**
1. $H_0: \mu = \mu_0$ $H_a: \mu \neq \mu_0$	If $t_0 > t_{n-1,\alpha/2}$ or $t_0 < -t_{n-1,\alpha/2}$, then reject H_0.
2. $H_0: \mu = \mu_0$ or $\mu \geq \mu_0$ $H_a: \mu < \mu_0$	If $t_0 < -t_{n-1,\alpha}$, then reject H_0.
3. $H_0: \mu = \mu_0$ or $\mu \leq \mu_0$ $H_a: \mu > \mu_0$	If $t_0 > t_{n-1,\alpha}$, then reject H_0.

Hypothesis Testing

Statistics

Example 2: It is claimed that the IQ scores of undergraduate students in our country have a normal distribution with a mean of 117. The IQ scores of 12 randomly selected students have been measured to research this claim for the mean. The following scores are obtained:

103, 96, 112, 130, 111, 125, 89, 100, 104, 117, 99, 118.

Test whether the population mean is 117 ($\alpha = 0.05$).

Hypothesis Testing

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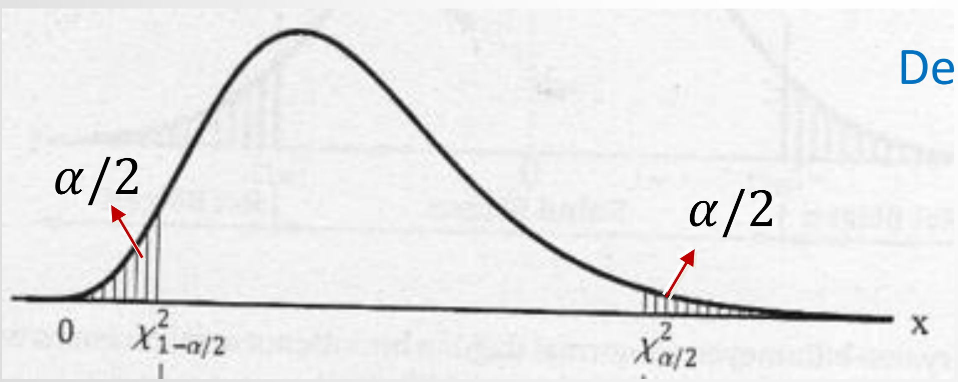
Hypothesis Testing for Population Variance σ^2 (for normally distributed populations)

1) Consider that $H_0: \sigma = \sigma_0$, $H_a: \sigma \neq \sigma_0$ for α.

Test Statistic: $\chi_0^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$

If H_0 is true, …



Decision Rule:

If $\chi^2_{n-1,1-\alpha/2} \leq \chi_0^2 \leq \chi^2_{n-1,\alpha/2}$, accept H_0. Otherwise, reject H_0.

$$\mathrm{P}\left(\chi^2_{n-1,1-\alpha/2} \leq \chi_0^2 \leq \chi^2_{n-1,\alpha/2}\right) = 1 - \alpha$$

The shaded-region is the rejection region, while other parts are the acceptance regions.

Hypothesis Testing

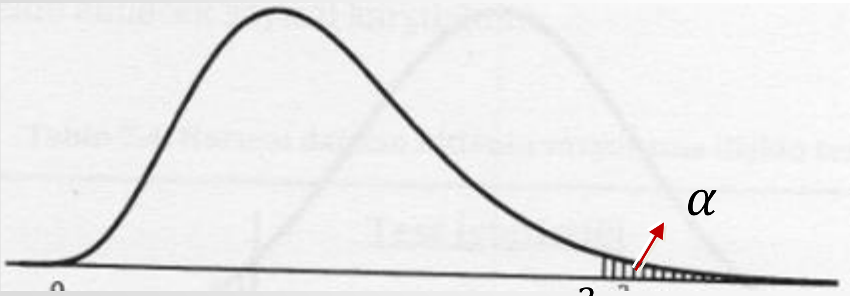
Statistics

Hypothesis Testing for Population Variance σ (for normally distributed populations)

2) Consider that $H_0: \sigma = \sigma_0$ or $\sigma \leq \sigma_0$ for α.
$H_a: \sigma > \sigma_0$

Test Statistic: $\chi_0^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$

If H_0 is true, ...



Decision Rule:

If $\chi_0^2 \leq \chi^2_{n-1,\alpha}$, accept H_0.
Otherwise, reject H_0.

$$\mathrm{P}\left(\chi_0^2 \leq \chi^2_{n-1,\alpha}\right) = 1 - \alpha$$

The shaded-region is the rejection region, while the other part is the acceptance region.

Hypothesis Testing

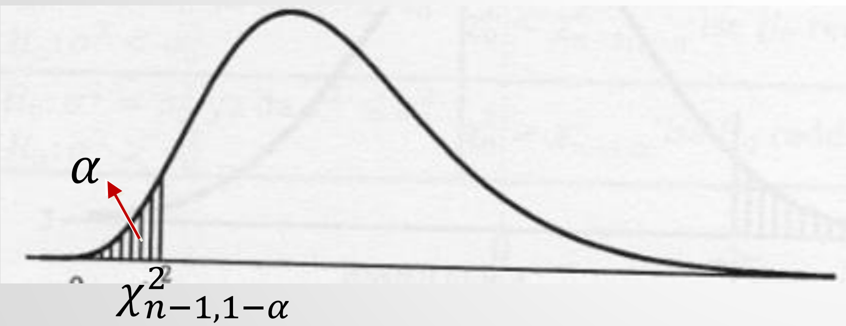
Statistics

Hypothesis Testing for Population Variance σ (for normally distributed populations)

3) Consider that $H_0: \sigma = \sigma_0$ or $\sigma \geq \sigma_0$ for α. Test Statistic: $\chi_0^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$

$H_a: \sigma < \sigma_0$

If H_0 is true, …



Decision Rule:

If $\chi^2_{n-1,1-\alpha} \leq \chi_0^2$, accept H_0.
Otherwise, reject H_0.

$$\mathrm{P}\left(\chi^2_{n-1,1-\alpha} \leq \chi_0^2\right) = 1 - \alpha$$

The shaded-region is the rejection region, while the other part is the acceptance region.

Hypothesis Testing

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Example 3: The crankshafts whose diameter are 4 cm are produced in a factory. The administrators of the factory do not want that the variance of diameters is greater than 0.25 cm in order to protect their production standards. If the variance is greater than the specified value, they will reorganize the production procedure. A sample of size 20 crankshafts is randomly selected. If the diameter of the crankshafts is assumed to be normally distributed and the sample variance is 0.32 cm, is a new production procedure required or not? ($\alpha = 0.05$).

Hypothesis Testing

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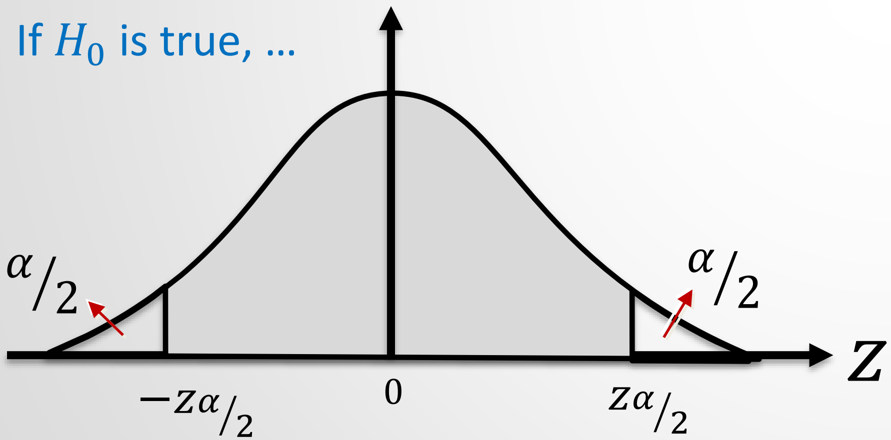
Hypothesis Testing for Population Proportion p ($n \geq 30$)

1) Consider that $H_0: p = p_0$, $H_a: p \neq p_0$ for α.

Test Statistic: $z_0 = \dfrac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$

If H_0 is true, …



Decision Rule:

If $-z_{\alpha/2} \leq Z_0 \leq z_{\alpha/2}$, accept H_0.
Otherwise, reject H_0.

$$P\left(-z_{\alpha/2} \leq Z_0 \leq z_{\alpha/2}\right) = 1 - \alpha$$

The shaded-region is the rejection region, while other parts are the acceptance regions.

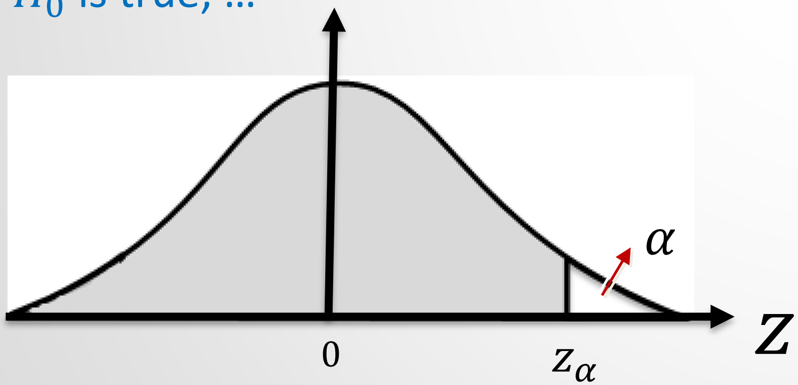
Hypothesis Testing

Statistics

Hypothesis Testing for Population Proportion p $(n \geq 30)$

2) Consider that $H_0: p = p_0 \ or \ p \leq p_0$, $H_a: p > p_0$ for α. Test Statistic: $z_0 = \dfrac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$

If H_0 is true, …



$$\mathrm{P}(Z_0 \leq z_\alpha) = 1 - \alpha$$

Decision Rule:

If $Z_0 \leq z_\alpha$, accept H_0.
Otherwise, reject H_0.

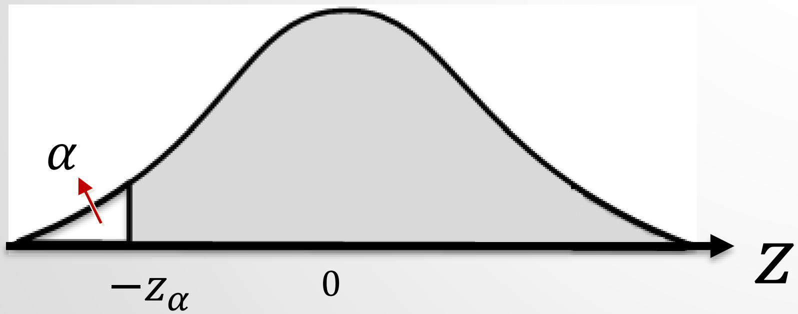
Hypothesis Testing

Statistics

Hypothesis Testing for Population Proportion p ($n \geq 30$)

3) Consider that $H_0: p = p_0 \; or \; p \geq p_0$, $H_a: p < p_0$ for α. Test Statistic: $z_0 = \dfrac{\hat{p} - p_0}{\sqrt{p_0 q_0 / n}}$

If H_0 is true, …



$$\mathrm{P}(-z_\alpha \leq Z_0) = 1 - \alpha$$

Decision Rule:

If $-z_\alpha \leq Z_0$, accept H_0.
Otherwise, reject H_0.

Hypothesis Testing

Statistics

Example 4: A drug company claimed that a newly produced drug has a side effect on at most 10% of the patients. The drug is tested on 300 patients, and it is observed that the side effect appeared in 45 patients. Is it very significant proof to reject the company's claim?

Remark: If a hypothesis is claimed for $\alpha = 0.01$, we say that it is very significant proof.
If α is selected as 0.05, we say that it is significant proof.

Hypothesis Testing

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Hypothesis Testing for the Difference of Population Means $(\boldsymbol{\mu_1 - \mu_2})$

A) For Normal Distributed Populations with Known Variance

To compare mean of the two populations, we can use the following hypothesis:

$$H_0: \mu_1 = \mu_2 \qquad \text{or} \qquad H_0: \mu_1 - \mu_2 = 0$$
$$H_a: \mu_1 \neq \mu_2 \qquad \phantom{\text{or}} \qquad H_a: \mu_1 - \mu_2 \neq 0$$

For more general analysis, we can use

$$H_0: \mu_1 - \mu_2 = \delta_0$$
$$H_a: \mu_1 - \mu_2 \neq \delta_0$$

Hypothesis Testing

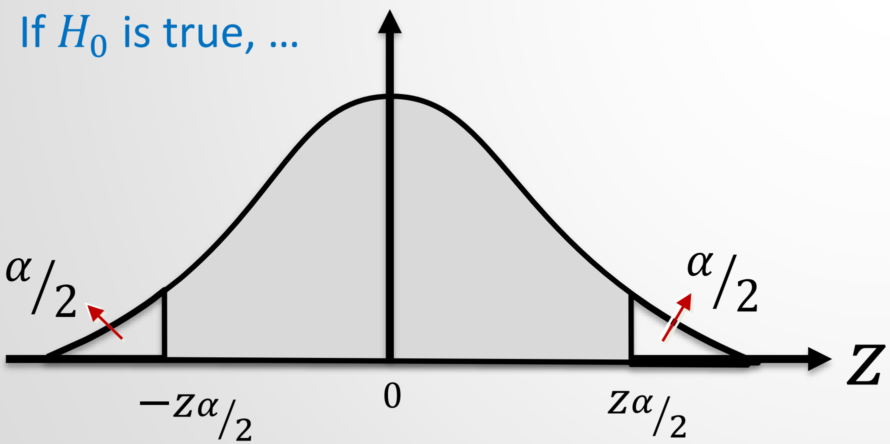
Statistics

Hypothesis Testing for the Difference of Population Means $(\mu_1 - \mu_2)$

1) Consider that $H_0: \mu_1 - \mu_2 = \delta_0$, $H_a: \mu_1 - \mu_2 \neq \delta_0$ for α.

Test Statistic: $Z_0 = \dfrac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$

If H_0 is true, …



$$P\left(-z_{\alpha/2} \leq Z_0 \leq z_{\alpha/2}\right) = 1 - \alpha$$

Decision Rule:

If $-z_{\alpha/2} \leq Z_0 \leq z_{\alpha/2}$, accept H_0.
Otherwise, reject H_0.

The gray-shaded region is the acceptance region, other parts are rejection regions.

Hypothesis Testing

Statistics

As seen, a hypothesis test can be performed like the mean of a single population. Thus, we can derive the following test statistics and decision rules similarly.

Test Statistic	Hypothesis testing for the difference of population mean $(\mu_1 - \mu_2)$
$z_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{\dfrac{{\sigma_1}^2}{n_1} + \dfrac{{\sigma_2}^2}{n_2}}}$	For the Normal distribution and known populations variances
$z_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{\dfrac{{\sigma_1}^2}{n_1} + \dfrac{{\sigma_2}^2}{n_2}}}$	For the unknown distribution ($n_1 \geq 30$ and $n_2 \geq 30$) and known populations variances
$z_0 = \dfrac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{\dfrac{{s_1}^2}{n_1} + \dfrac{{s_2}^2}{n_2}}}$	For the unknown distribution ($n_1 \geq 30$ and $n_2 \geq 30$) and unknown populations variances
Hypothesis	**Decision Rule**
1. $H_0: \mu_1 - \mu_2 = \delta_0$ $H_a: \mu_1 - \mu_2 \neq \delta_0$	If $z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$, then reject H_0.
2. $H_0: \mu_1 - \mu_2 = \delta_0$ or $\mu_1 - \mu_2 \geq \delta_0$ $H_a: \mu_1 - \mu_2 < \delta_0$	If $z_0 < -z_\alpha$, then reject H_0.
3. $H_0: \mu_1 - \mu_2 = \delta_0$ or $\mu_1 - \mu_2 \leq \delta_0$ $H_a: \mu_1 - \mu_2 > \delta_0$	If $z_0 > z_\alpha$, then reject H_0.

Hypothesis Testing

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$$t_0 = \frac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{S_k\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$ $$\upsilon = n_1 + n_2 - 2$$	For the Normal distribution and unknown populations variances $(\sigma_1{}^2 = \sigma_2{}^2)$
$$t_0 = \frac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{\frac{s_1{}^2}{n_1} + \frac{s_2{}^2}{n_2}}}$$ $$\upsilon = \frac{\left(\frac{s_1{}^2}{n_1} + \frac{s_2{}^2}{n_2}\right)^2}{\frac{(s_1{}^2/n_1)^2}{n_1 - 1} + \frac{(s_2{}^2/n_2)^2}{n_2 - 1}}$$	For the Normal distribution and unknown populations variances $(\sigma_1{}^2 \neq \sigma_2{}^2)$
Hypothesis	**Decision Rule**
1. $H_0: \mu_1 - \mu_2 = \delta_0$ $H_a: \mu_1 - \mu_2 \neq \delta_0$	If $t_0 > t_{\upsilon,\alpha/2}$ or $t_0 < -t_{\upsilon,\alpha/2}$, then reject H_0.
2. $H_0: \mu_1 - \mu_2 = \delta_0$ or $\mu_1 - \mu_2 \geq \delta_0$ $H_a: \mu_1 - \mu_2 < \delta_0$	If $t_0 < -t_{\upsilon,\alpha}$, then reject H_0.
3. $H_0: \mu_1 - \mu_2 = \delta_0$ or $\mu_1 - \mu_2 \leq \delta_0$ $H_a: \mu_1 - \mu_2 > \delta_0$	If $t_0 > t_{\upsilon,\alpha}$, then reject H_0.

Hypothesis Testing

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Example 5: A sample of 50 cigarettes in the first type has a mean of 2.61 mg and a standard deviation of 0.12 mg as well as a sample of 40 cigarettes in the second type has a mean of 2.38 mg and a standard deviation of 0.14 mg. Compare the hypothesis $\mu_1 - \mu_2 = 0.20$ to the alternative hypothesis $\mu_1 - \mu_2 \neq 0.20$ in the level of significance 0.05.

Hypothesis Testing

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Example 6: To compare two diet programs, a group of 9 people applied for the first diet program, and another group of 12 people applied for the second program during a month. It is observed that the people in the first group and the second group have lost an average of 7.2 kg and 6.9 kg, respectively. Their standard deviations are measured as 0.59 kg and 0.82 kg, respectively. By assuming that the population distributions are normally distributed, for $\alpha=0.05$, test the hypothesis "there is no significant difference between diet programs" with the alternative hypothesis "the first diet program is more efficient".

Hypothesis Testing

Statistics

Hypothesis Testing for the Difference of Population Proportions $(p_1 - p_2)$ $(n_1 \geq 30, n_2 \geq 30)$

To compare the proportion of the two populations, we can use the following hypothesis:

$$H_0: p_1 - p_2 = \delta_0$$
$$H_a: p_1 - p_2 \neq \delta_0$$

If $n_1 \geq 30$ and $n_2 \geq 30$, we know that the following random variable has approximately standard normal distribution:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$$

If we use weighted means $\hat{p}_w$ and $\hat{q}_w$ instead of $\hat{p}_1, \hat{q}_1, \hat{p}_2, \hat{q}_2$, the following test statistic is used:

$$Z_0 = \frac{(\hat{p}_1 - \hat{p}_2) - \delta_0}{\sqrt{\hat{p}_w\hat{q}_w\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, \qquad \hat{p}_w = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}, \qquad \hat{q}_w = 1 - \hat{p}_w$$

Hypothesis Testing

Statistics

We can derive the following test statistics and decision rules similarly.

Test Statistic	Hypothesis testing for the difference of the population proportion $(p_1 - p_2)$
$z_0 = \dfrac{(\hat{p}_1 - \hat{p}_2) - p_0}{\sqrt{\hat{p}_w \hat{q}_w \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	For ($n_1 \geq 30$ and $n_2 \geq 30$)
Hypothesis	**Decision Rule**
1. $H_0: p_1 - p_2 = \delta_0$ $H_a: p_1 - p_2 \neq \delta_0$	If $z_0 > z_{\alpha/2}$ or $z_0 < -z_{\alpha/2}$, then reject H_0.
2. $H_0: p_1 - p_2 = \delta_0$ or $p_1 - p_2 \geq \delta_0$ $H_a: p_1 - p_2 < \delta_0$	If $z_0 < -z_\alpha$, then reject H_0.
3. $H_0: p_1 - p_2 = \delta_0$ or $p_1 - p_2 \leq \delta_0$ $H_a: p_1 - p_2 > \delta_0$	If $z_0 > z_\alpha$, then reject H_0.

Hypothesis Testing

Statistics

Example 7: It is seen that 21 families have a villa in 220 families which are randomly selected in Istanbul, and 18 families have a villa in 177 families which are randomly selected in Ankara. Is there a significant difference between the proportions for the two cities regarding having a villa?

Hypothesis Testing

Statistics

Hypothesis Testing for the Rate of Population Variances $\left({}^{\sigma_1^2}\!/_{\sigma_2^2}\right)$

Assume that both populations have a normal distribution. We can use the following hypothesis:

$$H_0: \sigma_1^2 = \sigma_2^2 \qquad H_0: {}^{\sigma_1^2}\!/_{\sigma_2^2} = 1$$

$$H_a: \sigma_1^2 \neq \sigma_2^2 \quad \text{or} \quad H_a: {}^{\sigma_1^2}\!/_{\sigma_2^2} \neq 1$$

We know that

$$F_{n_1-1,n_2-1} = \frac{S_1^2 \sigma_2^2}{S_2^2 \sigma_1^2}$$

When H_0 is true ($\sigma_1^2 = \sigma_2^2$), the test statistic becomes:

$$F_0 = \frac{S_1^2}{S_2^2}$$

Hypothesis Testing

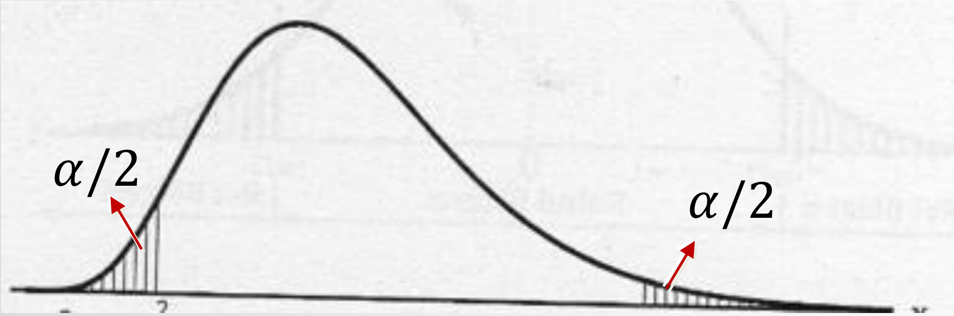
Statistics

Hypothesis Testing for the Rate of Population Variances (σ_1^2/σ_2^2) (for normal distributed populations)

Consider that $H_0: \sigma_1^2/\sigma_2^2 = 1$, $H_a: \sigma_1^2/\sigma_2^2 \neq 1$ for α.

Test Statistic: $F_0 = \dfrac{S_1^2}{S_2^2}$

If H_0 is true, …



$F_{n_1-1,n_2-1,1-\alpha/2}$ $F_{n_1-1,n_2-1,\alpha/2}$

Decision Rule:

If $F_{n_1-1,n_2-1,1-\alpha/2} \leq F_0 \leq F_{n_1-1,n_2-1,\alpha/2}$, accept H_0. Otherwise, reject H_0.

$$\mathrm{P}\left(F_{n_1-1,n_2-1,1-\alpha/2} \leq F_0 \leq F_{n_1-1,n_2-1,\alpha/2}\right) = 1-\alpha$$

The shaded regions are rejection regions, the other part is the acceptance region.

Hypothesis Testing

Statistics

We can derive the following test statistics and decision rules similarly.

Test Statistic	Hypothesis testing for the rate of populations variances $(\sigma_1{}^2/\sigma_2{}^2)$
$F_0 = \dfrac{S_1{}^2}{S_2{}^2}$	For the Normal distribution
Hypothesis	**Decision Rule**
1. $H_0: \sigma_1{}^2/\sigma_2{}^2 = 1$ $H_a: \sigma_1{}^2/\sigma_2{}^2 \neq 1$	If $F_0 > F_{n_1-1, n_2-1, \alpha/2}$ or $F_0 < F_{n_1-1, n_2-1, 1-\alpha/2}$, then reject H_0.
2. $H_0: \sigma_1{}^2/\sigma_2{}^2 = 1$ or $\sigma_1{}^2/\sigma_2{}^2 \geq 1$ $H_a: \sigma_1{}^2/\sigma_2{}^2 < 1$	If $F_0 < F_{n_1-1,\ n_2-1,\ 1-\alpha}$, then reject H_0.
3. $H_0: \sigma_1{}^2/\sigma_2{}^2 = 1$ or $\sigma_1{}^2/\sigma_2{}^2 \leq 1$ $H_a: \sigma_1{}^2/\sigma_2{}^2 > 1$	If $F_0 > F_{n_1-1,\ n_2-1,\ \alpha}$, then reject H_0.

Hypothesis Testing

Statistics

Example 8: In a soap factory, two machines fill the boxes whose capacities are 300 gr with soap powder. The randomly selected two samples of 10 boxes that are filled by these machines are weighted. If their standard deviations are 19 gr and 12 gr, respectively, test the hypothesis “there is no significant difference between two machines regarding the variance” with the alternative hypothesis “the first machine has more variance”. ($\alpha = 0.05$)

Hypothesis Testing

Statistics

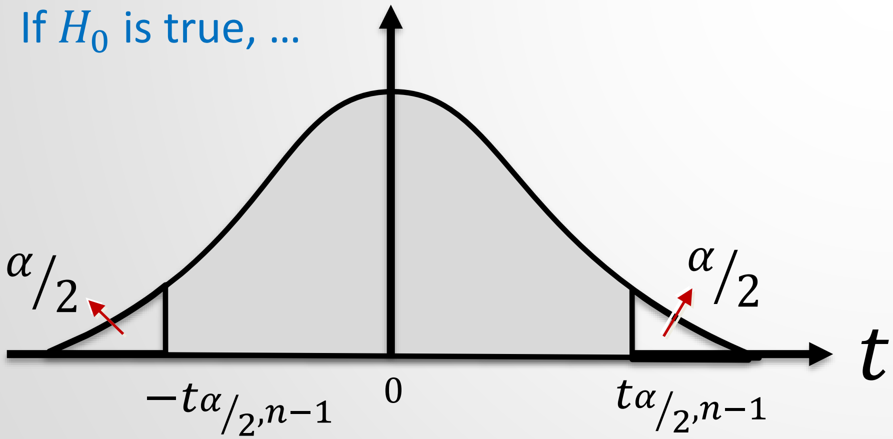
Hypothesis Testing for the Means of Paired Populations (μ_D)

A) For Normal Distributed Populations

1) Consider that $\begin{matrix} H_0: \mu_D = \delta_0 \\ H_a: \mu_D \neq \delta_0 \end{matrix}$ for α.

Test Statistic: $t_0 = \dfrac{\overline{D} - \delta_0}{\frac{S_D}{\sqrt{n}}}$

If H_0 is true, ...



Decision Rule:

If $-t_{\alpha/2,n-1} \leq t_0 \leq t_{\alpha/2,n-1}$, accept H_0. Otherwise, reject H_0.

$$P\left(-t_{\alpha/2,n-1} \leq t_0 \leq t_{\alpha/2,n-1}\right) = 1 - \alpha$$

The gray-shaded region is the acceptance region, other parts are rejection regions.

Hypothesis Testing

Statistics

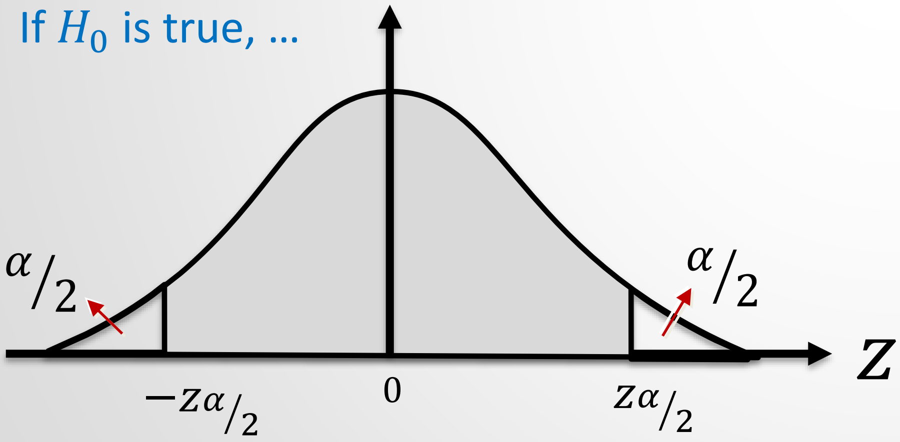
Hypothesis Testing for the Means of Paired Populations (μ_D)

B) For Any Populations, $n \geq 30$

1) Consider that $H_0: \mu_D = \delta_0$, $H_a: \mu_D \neq \delta_0$ for α.

Test Statistic: $z_0 = \dfrac{\overline{D} - \delta_0}{\frac{S_D}{\sqrt{n}}}$

If H_0 is true, ...



$$\mathrm{P}\left(-z\alpha/_2 \leq z_0 \leq t\alpha/_2\right) = 1 - \alpha$$

Decision Rule:

If $-z\alpha/_2 \leq z_0 \leq t\alpha/_2$, accept H_0.
Otherwise, reject H_0.

The gray-shaded region is the acceptance region, other parts are rejection regions.

Hypothesis Testing

Statistics

We can derive the following test statistics and decision rules similarly.

Test Statistic	Hypothesis testing for the means of paired populations (μ_D)
$t_0 = \dfrac{\bar{D} - \delta_0}{S_D / \sqrt{n}}$	For a normal distribution
Hypothesis	**Decision Rule**
1. $H_0: \mu_D = \delta_0$ $H_a: \mu_D \neq \delta_0$	If $t_0 > t_{\alpha/2, n-1}$ or $t_0 < -t_{\alpha/2, n-1}$, then reject H_0
2. $H_0: \mu_D = \delta_0$ or $\mu_D \geq \delta_0$ $H_a: \mu_D < \delta_0$	If $t_0 < -t_{\alpha, n-1}$, then reject H_0
3. $H_0: \mu_D = \delta_0$ or $\mu_D \leq \delta_0$ $H_a: \mu_D > \delta_0$	If $t_0 > t_{\alpha, n-1}$, then reject H_0
$Z_0 = \dfrac{\bar{D} - \delta_0}{S_D / \sqrt{n}}$	For any distribution ($n \geq 30$)
Hypothesis	**Decision Rule**
1. $H_0: \mu_D = \delta_0$ $H_a: \mu_D \neq \delta_0$	If $Z_0 > Z_{\alpha/2}$ or $Z_0 < -Z_{\alpha/2}$, then reject H_0
2. $H_0: \mu_D = \delta_0$ or $\mu_D \geq \delta_0$ $H_a: \mu_D < \delta_0$	If $Z_0 < -Z_{\alpha}$, then reject H_0
3. $H_0: \mu_D = \delta_0$ or $\mu_D \leq \delta_0$ $H_a: \mu_D > \delta_0$	If $Z_0 > Z_{\alpha}$, then reject H_0

Hypothesis Testing

Statistics

Example 9: To test the effect of an online tutorial on the students, 10 students are randomly chosen and their grades are measured before and after the online tutorial.

Students	After the online tutorial	Before the online tutorial
Student_1	62	61
Student_2	71	76
Student_3	66	70
Student_4	87	91
Student_5	69	46
Student_6	77	80
Student_7	45	37
Student_8	61	73
Student_9	89	90
Student_10	65	70

Test the hypothesis "the online tutorial increases the student grades" with significance level $\alpha = 0.05$. (Assume that populations have normal distribution.)

Hypothesis Testing

Statistics

Students	X_{1i}	X_{2i}	$D_i = X_{1i} - X_{2i}$
Student_1	62	61	1
Student_2	71	76	-5
Student_3	66	70	-4
Student_4	87	91	-4
Student_5	69	46	23
Student_6	77	80	-3
Student_7	45	37	8
Student_8	61	73	-12
Student_9	89	90	-1
Student_10	65	70	-5

$$H_0: \mu_D \leq 0$$

$$H_a: \mu_D > 0$$

$$\bar{D} = \frac{\sum_{i=1}^{n} D_i}{n} = -0.2$$

$$S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \bar{D})^2}{n-1}} \cong 9.60$$

$$t_0 = \frac{\bar{D} - \delta_0}{S_D/\sqrt{n}} = \frac{-0.2 - 0}{9.60/\sqrt{10}} \cong -0.066$$

$$t_{n-1,\alpha} = t_{10-1,0.05} = 1.833$$

Since $t_0 \leq t_{n-1,\alpha}$, we accept H_0. The online tutorials have not increased the grades.

Hypothesis Testing

Statistics

p-value

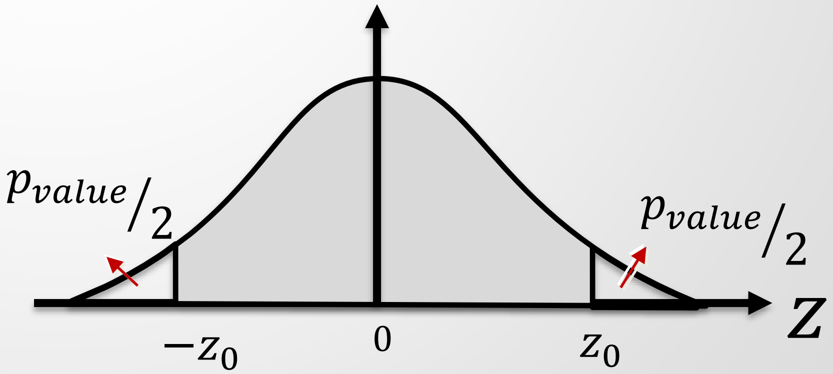
Consider a two-tailed hypothesis testing. *p*-value is a probability which can be determined from the following expression:

$$P(-z_0 \leq z \leq z_0) = 1 - p\_value$$

where z_0 is the computed test statistic. We can also make a decision about hypothesis testing by using the *p*-value:

If $p_{value} < \alpha$ ⇒ Reject H_0

If $p_{value} \geq \alpha$ ⇒ Accept H_0



Hypothesis Testing

Statistics

p-value

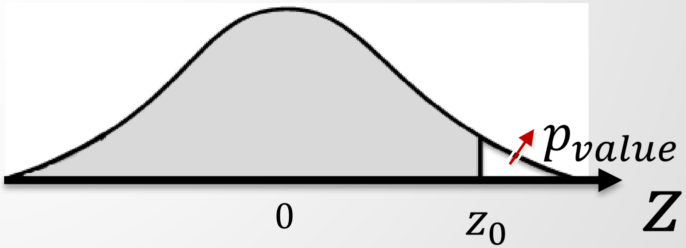
For one-tailed hypothesis testing, the *p*-value can be determined from the following expressions:

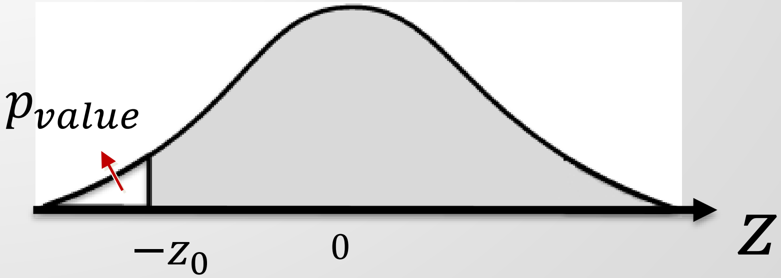
$$P(z \leq z_0) = 1 - p\_value \qquad P(-Z_0 \leq z) = 1 - p\_value$$

where z_0 is the computed test statistic. We can also a make decision about hypothesis testing by using the *p*-value:

If $p_{value} < \alpha$ → Reject H_0

If $p_{value} \geq \alpha$ → Accept H_0





Hypothesis Testing

Statistics

Example 10: Find p values for Ex1, Ex2 and Ex 4.